

Multi-Key Homomorphic Signatures for Approximate Computation:

Towards Verifiable Federated Authentication

PhD Research Proposal
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Post-Quantum Cryptography · Lattice-Based Signatures · Verifiable Computation · Federated Learning

Abstract

Imagine ten hospitals training local machine learning models that wish to collaborate on a global model without sharing raw patient data. They send their updates to a central server, which aggregates the results and sends them back. The following question cannot be answered by any existing cryptographic tool: how can anyone verify that the aggregated result is indeed derived from the legitimately signed data of those hospitals, given that the updates are floating-point approximations? To fill this gap, this proposal presents **Approximate Multi-Key Homomorphic Signatures (ApxMKHS)**, a new primitive that provides multi-party authentication, approximate correctness (with tolerance ε), post-quantum security and context hiding at the same time. The goal is 3–4 publications at CRYPTO, EUROCRYPT, ASIACRYPT and IEEE S&P in 3.5 years.

Keywords: Homomorphic Signatures · Multi-Key · Approximate Computation · Lattice Cryptography · Federated Learning · Post-Quantum Security.

1 Introduction

In modern computation, data rarely comes from a single source. In federated machine learning, healthcare analytics, and cloud outsourcing, inputs are generated by many independent parties, processed by servers that the data owners do not control, and consumed by verifiers who may never see the originals. Fully Homomorphic Encryption (FHE) [22] protects privacy by enabling computation on encrypted data but provides no assurances on integrity — whether the result was honestly computed from legitimately authorised inputs.

Homomorphic Signatures (HS) [27] solve this problem by allowing an evaluator to compute a valid signature on output $y = f(m_1, \dots, m_k)$ from signatures on inputs m_i without any secret key. The field has moved from early linear schemes [3, 8, 10, 23] to polynomial and circuit-level evaluation [9, 20], and finally to Multi-Key HS (MKHS), which enables data from several independent signers [2, 18].

However, one assumption has remained throughout this whole development: for every HS scheme, the output that is evaluated must be exactly correct. If the verifier accepts signature σ^* on output y , and the actual function output is $y' \neq y$, then the verification fails — regardless of how small the difference. That’s no small limitation. It is a fundamental mismatch with what modern computation actually is. The CKKS scheme [14] is the most common method of FHE used in privacy-preserving machine learning. It deals with approximate arithmetic: its outputs are noisy (within bounds by design), and decryption yields a value ε close to the plaintext (not the plaintext itself). Federated learning works the same way: gradient

vectors are floating-point approximations, not integers, not exact. There is no exact correctness possible in either case. But in both cases, immediate authentication is needed.

2 Literature Survey

2.1 Single-key HS

Linear schemes were the first HS schemes introduced. The first pairing-based scheme was proposed by Boneh et al. [10]. Lattice-based linear HS under SIS were proposed by Boneh and Freeman [8]. Constructions in the standard model with growing privacy [3, 4] culminated in full context hiding using Groth-Sahai proofs. Structure-preserving [19, 30] and adaptively secure lattice [24] versions followed. Nonlinear evaluation was enabled by ideal lattice ring structure [9], followed by signature-size reduction [25]. The first fully leveled HS scheme was based on Homomorphic Trapdoor Functions [23]. Modern constructions follow a Commit-and-Prove blueprint [13, 21, 26], the first unlevelled FHS from falsifiable assumptions used indistinguishability obfuscation [20].

2.2 Multi-key HS

Fiore et al. [18] introduced MKHS, which was later shown to be vulnerable to insider corruption [32]. The Matrioska compiler [17] converts single-key HS into multi-key HS using NIZKs, with a signature size increase of $O(k)$. Insider-corruption was countered in [29]. Batch arguments (BARGs) and functional commitments enable Anthoine et al. [2] to achieve fully succinct MKHS under SIS/LWE. Context-hiding MKHS with multi-hop evaluation was achieved in [1]; forward unforgeability under key leakage in [32].

2.3 Approximate computation

The CKKS scheme [14] supports FHE over the reals with outputs in bounded ε and is a standard for privacy-preserving ML. Cascudo et al. [11] studied verifiable computation for approximate FHE, which is a solution for computational integrity but not input authenticity. Model poisoning [5] and Byzantine attacks [6] are well-known challenges for federated learning [31] while secure aggregation [7] guarantees confidentiality but not authenticity. The application is further motivated by the ICC's own work [28] on privacy-preserving federated learning in the medical diagnosis.

3 Problem Statement

Core Problem. *No cryptographic primitive currently provides all of: (i) multi-key authentication across independent signers; (ii) approximate correctness verification within tolerance ε ; (iii) post-quantum security under SIS/LWE; (iv) succinctness independent of the number of signers; and (v) context hiding. Existing MKHS satisfies (i), (iii), (iv) but fails on (ii). Approximate FHE satisfies (ii) with no authentication whatsoever.*

3.1 Knowledge Gaps

Taken together, the preceding review reveals five gaps, each of which must be addressed before ApxMKHS can exist as a well-founded primitive:

Gap 1 (Definitions):

The unforgeability games of Catalano et al. [12] and Fiore et al. [18] are undefined when outputs are equivalence classes up to ε rather than exact values. New security definitions are needed from scratch.

Gap 2 (Linear Construction):

No ApxMKHS scheme exists for any function class. How combined lattice norm growth and approximation error interact has not been studied; no tools exist for bounding their joint effect.

Gap 3 (Succinctness):

The BARG paradigm of Anthoine et al. [2] handles exact NP relations only. There are no known approximate BARGs or approximate functional commitments.

Gap 4 (Forward Unforgeability):

The forward unforgeability of Xu and Nishide [32] relies on exact verification equations. Its extension to approximate settings, where verification tolerates bounded deviations, has not been attempted.

Gap 5 (Application):

Despite clear practical need, no cryptographic framework exists for authenticated approximate aggregation in federated learning or CKKS-based computation.

These gaps translate into five research questions:

RQ1. Definitions.

How should unforgeability and context hiding be formally defined for MKHS when outputs are approximate? How can a security game parameterised by tolerance ε prevent an adversary from exploiting approximation to mask forgeries?

RQ2. Linear Construction.

Can a secure ApxMKHS scheme be constructed for linear functions under standard lattice assumptions (SIS/LWE)? Using ZK range arguments [16] to prove the output lies within tolerance ε without revealing the exact approximation error? How does combined noise (approximation error and lattice norm growth) constrain the number of admissible signers?

RQ3. Succinct Construction.

Can the BARG and Functional Commitment paradigm of Anthoine et al. [2] be extended to approximate settings, yielding a fully succinct ApxMKHS with constant signature size?

RQ4. Forward Unforgeability.

Can the forward unforgeability notion of Xu and Nishide [32] — preventing forgery even after signing key leakage — be preserved in the approximate setting?

RQ5. Application.

How can ApxMKHS be concretely instantiated for federated learning, and what are the practical parameter trade-offs for realistic ML workloads?

3.2 Contributions**C1. Definitional Framework:**

The first formal definitions of ε -approximate unforgeability, approximate context hiding, and forward unforgeability in the multi-key HS setting, together with separation results establishing their relative strength.

C2. Linear ApxMKHS:

The first construction for linear functions based on standard lattice (SIS/LWE) and pairing-based assumptions, with provable ε -approximate unforgeability.

C3. Succinct ApxMKHS:

A fully succinct scheme with constant-size evaluated signatures, achieved by extending the BARG and approximate functional commitment paradigm to the multi-key approximate setting.

C4. Application Framework:

A concrete instantiation for federated learning integrity, with evaluated parameters, prototype implementation, and formal security analysis.

4 Work Plan

Phase 1 (Year 1): Formal Definitions.

Formalise (ε, δ) -correctness, ε -approximate unforgeability, and approximate context hiding. Prove separation results showing approximate unforgeability is strictly weaker than its exact counterpart, and that new attack vectors arise from the tolerance region. **Target:** TCC/PKC 2027.

Phase 2 (Year 2): Linear ApxMKHS Construction.

Extend the forward-unforgeable MKHS of Xu and Nishide [32] to accept results that are approximately correct within tolerance ε , rather than exactly correct. The key technical finding — drawing on prior work in [23] and [18] — is that lattice signature noise and approximation error add together rather than compound, which tells us how many signers and operations the scheme can support. Proving that the output lies within tolerance without revealing the exact error uses the zero-knowledge range argument technique from the UOW group [16]. Security reduces to SIS. **Target:** ASIACRYPT/PKC 2028.

Phase 3 (Year 2.5–3): Succinct ApxMKHS.

Aggregate the k ZK range proofs into a single batched lattice argument exploiting SIS structure, targeting $O(\log k)$ evaluated signature size. Investigate approximate BARGs [15] and approximate functional commitments [13] as a route to full $O(1)$ succinctness. **Target:** EUROCRYPT/CRYPTO 2029.

Phase 4 (Year 3–3.5): Application and Implementation.

Instantiate ApxMKHS for the medical federated learning setting of [28], with ε derived from floating-point gradient precision. Implement in OpenFHE + lattice library; benchmark for 10–100 institutions. Prove security against model poisoning [5]. **Target:** IEEE S&P/CCS 2029.

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